

■ Computation of the reference density

Global counter GC:

total number of all references

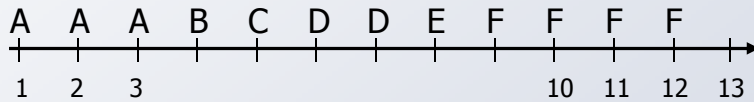
Arrival time AT:

GC value when the page enters the buffer

Reference counter RC

Reference density

$$RD(j) = \frac{RC(j)}{GC - AT(j)}$$

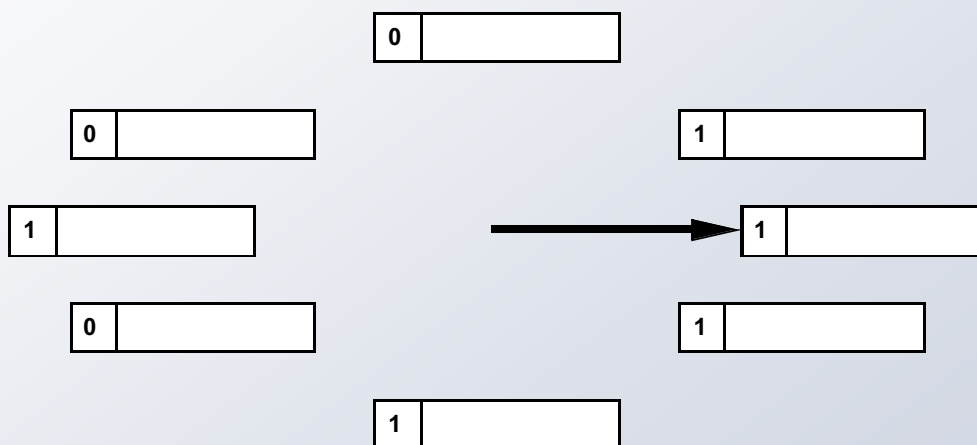


	RC	AT	RD
A	3	1	
B	1	4	
C	1	5	- - -
D	2	6	
E	1	8	
F	4	9	

Figure 1: Computation of RD (Credits: http://www.lgis.informatik.uni-kl.de/cms/fileadmin/courses/SS2011/RDBS/lectures/Chapter_04.BufferManagement.pdf)

Clock

- Clock is an efficient version of FIFO, where data in the cache does not have to be constantly pushed to the back of the list but performs a general function as 'Second Chance' in a circular queue.
- 'Second Chance' is a bit assigned to each object data in the cache, and it is set to be 1 when it has been referenced, giving it a second chance.
- During the eviction operation, the object follows the FIFO queue, but to be evicted, the object needs to be 0. So, if in the queue the object was recently accessed it would have been '1', though after the eviction the flag is reset to 0 again.
- Thus the data gets a second chance of not being replaced during its first consideration.



→ approximate consideration of the most recent reference time

Figure 2: Reference time (Credits: http://www.lgis.informatik.uni-kl.de/cms/fileadmin/courses/SS2011/RDBS/lectures/Chapter_04.BufferManagement.pdf)